CHAPTER 6

Divisibility and congruences

Divisibility in $\mathbb{Z}$, Euclidean division, congruences

Objectives. Use divisibility in $\mathbb{Z}$ and its properties; carry out a Euclidean division; compute with congruences; establish a divisibility test; study remainders of powers.

Prerequisites. Multiples, divisors, prime numbers (Seconde); proof by induction; integer part of a real number.

Fluency

Fluency · No calculator, five minutes.

1. List the positive divisors of 36.
2. Is the number $n(n+1)$ always even? Justify.
3. What is the units digit of 7^4?
4. Simplify the fraction $\frac{84}{126}$.
5. Is the number 2,026 divisible by 3? by 2?
6. Compute 17×23.